

例4 系统结构图如图1所示，给出三种侯选的串联校正装置 $G_c(s)$ ，其对数幅频特性分别如图2中所示。试问：

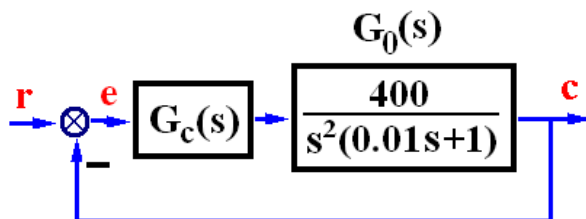


图1 系统结构图

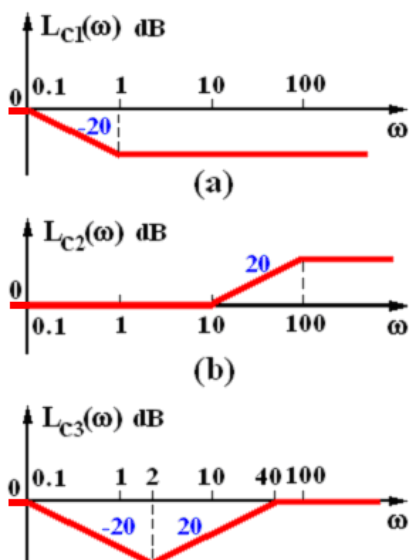
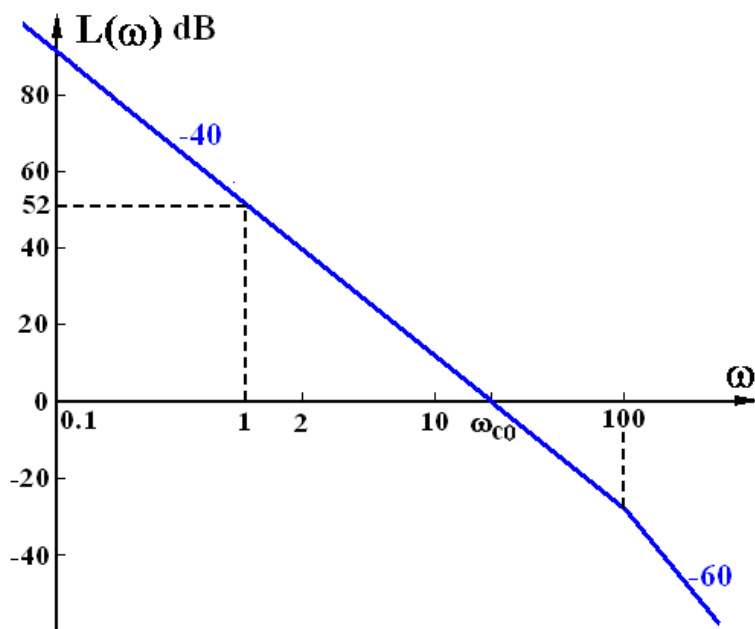


图2 候选校正装置对数幅频特性

- (1) 使用哪一种校正装置可使系统的稳定程度最好？
- (2) 欲将12 (Hz) 的正弦噪声削弱10倍左右，应采用哪种校正装置？

解，原系统开环传递函数为

$$G_0(s) = \frac{400}{s^2 \left(\frac{s}{100} + 1 \right)} \quad \begin{cases} K = 400 \\ v = 2 \end{cases}$$



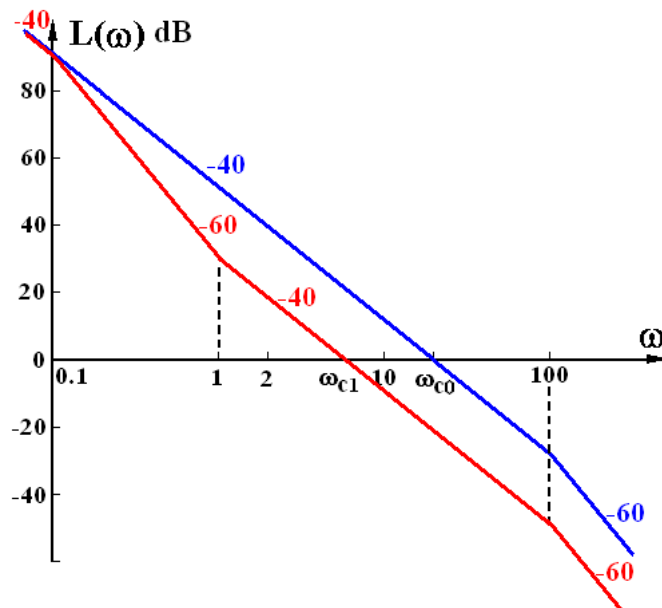
绘制 $L_0(\omega)$ ，可得：

$$\omega_{c0} = 20$$

$$\gamma_0 = 180^\circ - 180^\circ - \arctan \frac{20}{100} = -11.3^\circ \quad (\text{系统不稳定})$$

① 采用 $G_{c1}(s) = \frac{s+1}{10s+1}$

系统开环传递函数为 $G_{c1}(s) \cdot G_0(s) = \frac{400(s+1)}{s^2(0.01s+1)(10s+1)}$



画出对数幅频特性曲线 $L_{c1}(\omega)$ 可得：

$$G_{c1}(s) \cdot G_0(s) = \frac{400(s+1)}{s^2(0.01s+1)(10s+1)}$$

$$|G(j\omega_{c1})| = 1 \approx \frac{400\omega_{c1}}{\omega_{c1}^2 \times 1 \times 10\omega_{c1}} = \frac{40}{\omega_{c1}^2}$$

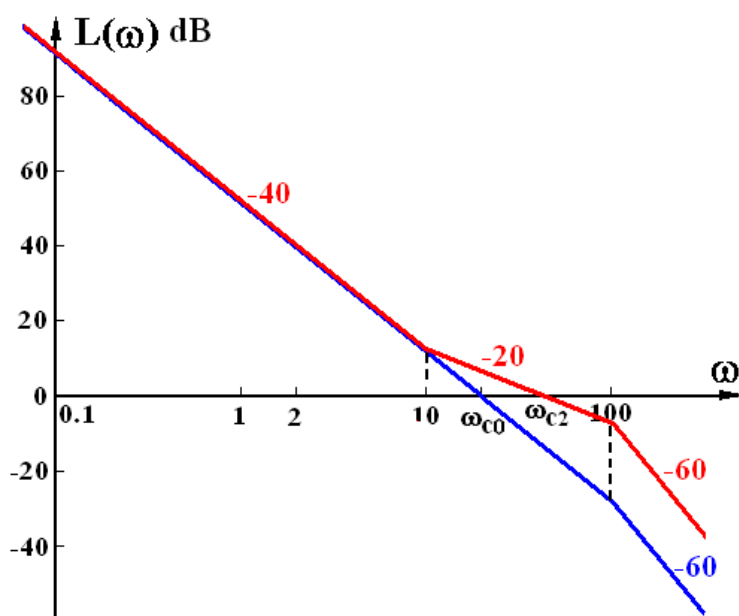
截止频率 $\omega_{c1} = \sqrt{40} = 6.32$

相角裕度 $\gamma_1 = 180^\circ + \varphi_1(\omega_{c1}) = -11.7^\circ$ (系统不稳定)

② 采用 $G_{c2}(s) = \frac{0.1s+1}{0.01s+1}$

系统开环传递函数为 $G_{c2}(s) \cdot G_0(s) = \frac{400(0.1s+1)}{s^2(0.01s+1)^2}$

画出对数幅频特性曲线 $L_2(\omega)$ 所示：



截止频率 $\omega_{c2} = \frac{\omega_0^2}{10} = \frac{20^2}{10} = 40$

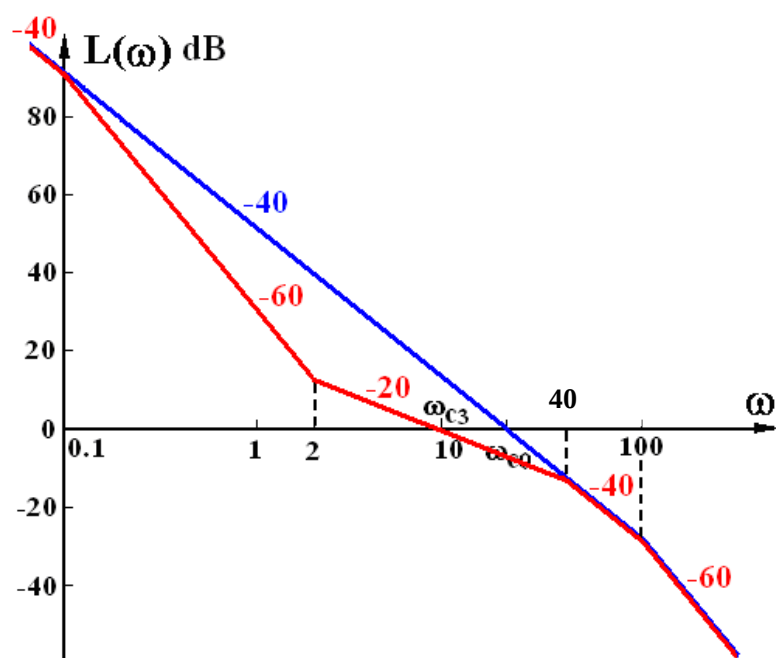
相角裕度 $\gamma_2 = 180^\circ + \varphi_2(\omega_{c2}) = 32.36^\circ$

③ 采用 $G_{c3}(s) = \frac{(0.5s + 1)^2}{(10s + 1)(0.02s + 1)}$

系统开环传递函数为

$$G_{c3}(s) \cdot G_0(s) = \frac{400(0.5s + 1)^2}{s^2(0.01s + 1)(10s + 1)(0.025s + 1)}$$

画出对数幅频特性曲线 $L_3(\omega)$ 所示：



截止频率 $\omega_{c3} = \frac{\omega_{c0}^2}{40} = \frac{20^2}{40} = 10$

相角裕度 $\gamma_3 = 180^\circ + \varphi_3(\omega_{c3}) = 48.21^\circ$

可见，采用 $G_{c1}(s) = \frac{s+1}{10s+1}$ 时系统不稳定；

采用 $G_{c3}(s) = \frac{(0.5s+1)^2}{(10s+1)(0.02s+1)}$ 时稳定程度最好。

(2) 确定使12Hz正弦噪声削弱10倍左右的校正网络

$$f = 12 \text{ Hz 时}, \quad \omega = 2\pi f = 75.4 \text{ (rad / s)}$$

对单位反馈系统，高频段闭环幅频特性与开环幅频特性基本一致。

从Bode图上看，在 $\omega = 75.4$ 处，有

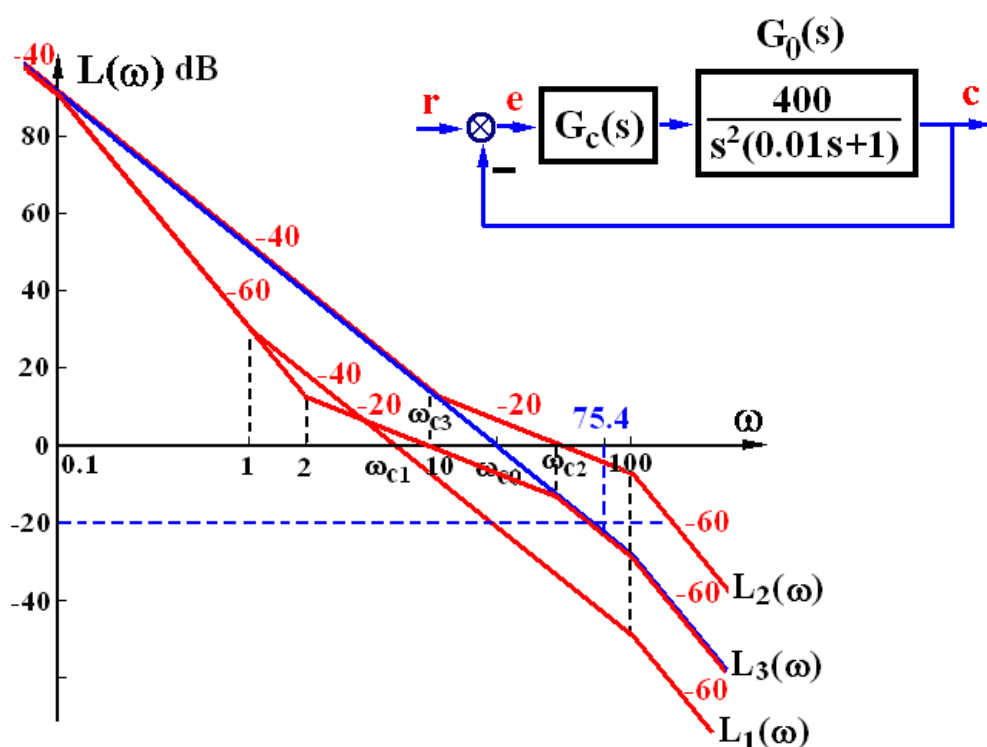
$$L_3(75.4) = -23\text{dB}$$

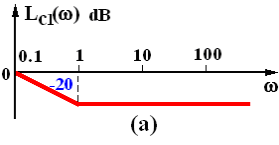
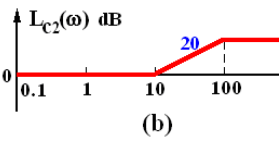
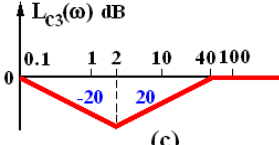
衰减倍数 $\alpha_3 = 10^{\frac{23}{20}} = 14.13 > 10$ ，可见，采用

$$G_{c3}(s) = \frac{(0.5s + 1)^2}{(10s + 1)(0.02s + 1)}$$

可以使 $12 \text{ Hz} = 75.4 \text{ rad/s}$ 的正弦信号

衰减14.13倍，满足系统要求。



$L_{ci}(\omega)$		 (a)	 (b)	 (c)
$G_{ci}(s)$	1	$\frac{s+1}{10s+1}$	$\frac{0.1s+1}{0.01s+1}$	$\frac{(0.5s+1)^2}{(10s+1)(0.02s+1)}$
ω_{ci} (rad/s)	20	6.32	40	10
γ_i	-11.3°	-11.7°	32.36°	48.21°
$L_{ci}(75.4)$ (dB)	-24dB (0.063)	-43dB (0.0071)	-5dB (0.56)	-24dB (0.063)